money and customer dissatisfaction. They may even affect one’s reputation, damaging the organization’s future market opportunities.

Whether there is a failure in the infrastructure or the software, your goal is to guarantee the availability and reliability of your systems, even if only in a degraded operational mode. To guarantee availability, you need to have something in place to face the failures when they arrive, handle them, and ensure that the overall system can still provide services to its users. Any action required to process the failure or tasks like upgrades should require zero downtime. Customers expect that.

We want cloud native applications to be resilient, and cloud technologies provide strategies for implementing resilient infrastructure. If being always available, safe, and resilient is a requirement for your business, the cloud native approach is a good option for you. The resilience of a software system, in turn, enables speed: the more stable the system, the more frequently you can safely release new features.

1.5.3 Scale

Elasticity is about being able to scale your software depending on the load. You can scale an elastic system to ensure an adequate service level for all your customers. If there is a higher load than usual, you’ll need to spin up more instances of your services to support that extra traffic. Or perhaps something terrible happened and some services failed—you’ll need to be able to spin up new instances to replace them.

Foreseeing what will happen is hard, if not impossible. It’s not enough to build scalable applications—you need them to scale dynamically. Whenever there is a high load, your system should scale out dynamically, quickly, and painlessly. When the high peak is over, it should scale in again.